

8.6

Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males and females yields $\hat{\sigma}_M^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

c. Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared residuals is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- d. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- e. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

- f. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{ccccccc} \widehat{WAGE} = & -17.77 & + 2.50EDUC & + 0.23EXPER & + 3.23METRO & - 4.20FEMALE & \\ (se) & (2.36) & (0.14) & (0.031) & (1.05) & (0.81) & \\ (robse) & (2.50) & (0.16) & (0.029) & (0.84) & (0.80) & \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results? $EDUC$ other narrower
f. If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

a. $H_0: \sigma_M^2 = \sigma_F^2$ vs. $H_1: \sigma_M^2 \neq \sigma_F^2$ # of males: 577, # of females: 423

$$\hat{\sigma}_M^2 = \frac{\sum e_i^2}{577-4} = \frac{97161.9174}{573} = 169.5670 \quad \text{Rejection Region: } R = \{F: F > F_{573,419}(0.975) = 1.1969 \text{ or } F < F_{573,419}(0.025) = 0.83777\}$$

By Goldfeld-Quandt test, the test statistic is $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.5670}{(12.024)^2} = 1.1729 \sim F_{573,419} \therefore F \notin R$

$\therefore$ We don't have enough evidence to reject $\sigma_1^2 = \sigma_2^2$, we can conclude that variance of male and female are the same

b. $H_0: \sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ $H_1: \sigma_{SINGLE}^2 < \sigma_{MARRIED}^2$ # of singles: 400, # of married: 600

$$\hat{\sigma}_S^2 = \frac{56231.0382}{400-5} = 142.3571 \quad \sigma_F^2 = \frac{100703.0471}{600-5} = 169.2488 \quad \text{Rejection Region: } R = \{F: F < F_{395,595}(0.05) = 0.85863\}$$

By Goldfeld-Quandt test, the test statistic is $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{142.3571}{169.2488} = 0.8411 \sim F_{395,595} \notin R$

$\therefore$ We have enough evidence to reject $\sigma_1^2 = \sigma_2^2$, we can conclude that variance of married and single are NOT the same

c. Rejection Region $R = \{\chi^2: \chi^2 > \chi_{0.95}^2(4) = 9.48777\}$

Test Statistic: $\chi^2 = 59.03 \sim \chi^2(4) \in R$

$\therefore$ We have enough evidence to reject $\sigma_1^2 = \sigma_2^2$, we can conclude the variance of married and single are NOT the same
It provides evidence about the issue discussed in (b), error variation is different for married and single.

d. White test: consider $x_i, x_i^2, x_i x_j, i, j = 1, 2, 3, 4, i \neq j \Rightarrow df: 4 + 4 + C_2^4 = 14$

Rejection Region $R = \{\chi^2: \chi^2 > \chi_{0.95}^2(14) = 23.6848\}$

Test Statistic: $\chi^2 = 78.82 \sim \chi^2(14) \in R$

$\therefore$ We have enough evidence to reject $\sigma_1^2 = \sigma_2^2$, we can conclude the variance of married and single are NOT the same
It provides evidence about the issue discussed in (b), error variation is different for married and single.

e. Interval with usual standard error: $[\beta_i \pm se(\beta_i)] \Rightarrow \text{range} = 2 se(\beta_i)$

Interval with robust standard error: $[\beta_i \pm robse(\beta_i)] \Rightarrow \text{range} = 2 robse(\beta_i)$

$\Rightarrow$ Narrower interval for robust error: EXPER, METRO, FEMALE

Wider interval for robust error: constant, EDUC

There isn't any inconsistency, since heteroskedasticity changes the standard errors but not change the OLS coefficient estimates
Usual SE 可能不准, 因此用 robust se 解决

f. It is compatible with the result (b). Because (b) tests whether the variance differs between married and single individuals, while (f) affects whether being married affects the mean wage. It is possible that both cases happen.

8.8.2 Computer Exercises

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

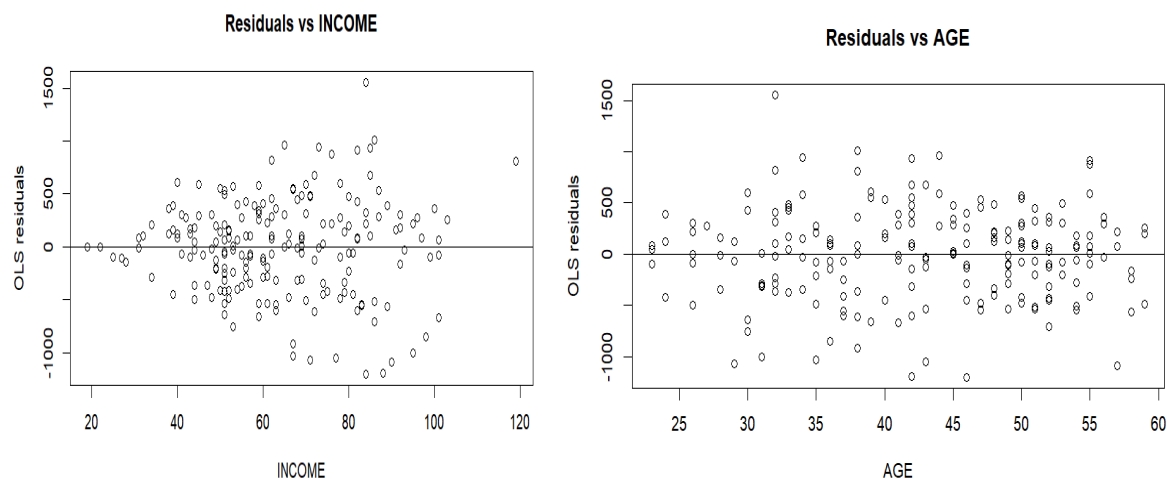
8.17 In this exercise, we explore the relationship between total household expenditures and expenditures on (a) more observations) and observations for which

a. The interval estimate is defined as:

$$[b_i \mp t_{\alpha/2} se(b_i)] \dots (1)$$

By running the code, we have the 95% confidence interval: [-135.3298 , -28.32302]

b. The residual plots are shown as below



We found that residual varies the same across ages, and the spread of the residual gets wider as income increases.

c. $H_0: \sigma_{small90}^2 = \sigma_{large90}^2$ v.s. $H_1: \sigma_{small90}^2 \neq \sigma_{large90}^2$ with $\alpha = 0.05$

We found that the test statistic is $\frac{\widehat{\sigma_{small90}^2}}{\widehat{\sigma_{large90}^2}} = 0.3221587$, and the rejection region is $\{F:$

$F < 0.6534355$ or $F > 1.530373\}$. Since the test statistic is in the critical region, we can

conclude that the variance of the 2 groups are not the same. The result is also consistent with (b).

d. The interval estimate is defined in (1), replacing $se(b_i)$ with robust standard error of b_i , and we have the result $[-139.323, -24.32986]$, we found that the confidence interval is wider than (a), to obtain more reliable inference under heteroskedasticity

e. Because $Var(e_i) = \sigma^2 \text{income}_i^2$, dividing the model by income_i makes the transformed error homoskedastic, allowing GLS estimation. By calculation in (a) and (d), we have the following confidence interval:

Conventional GLS: $[-119.8945, -33.71808]$

Robust GLS: $[-121.4134, -32.19919]$

Compare to (a), both conventional and robust GLS have narrower confidence interval, satisfying that GLS is have more precise result and narrower confidence interval.